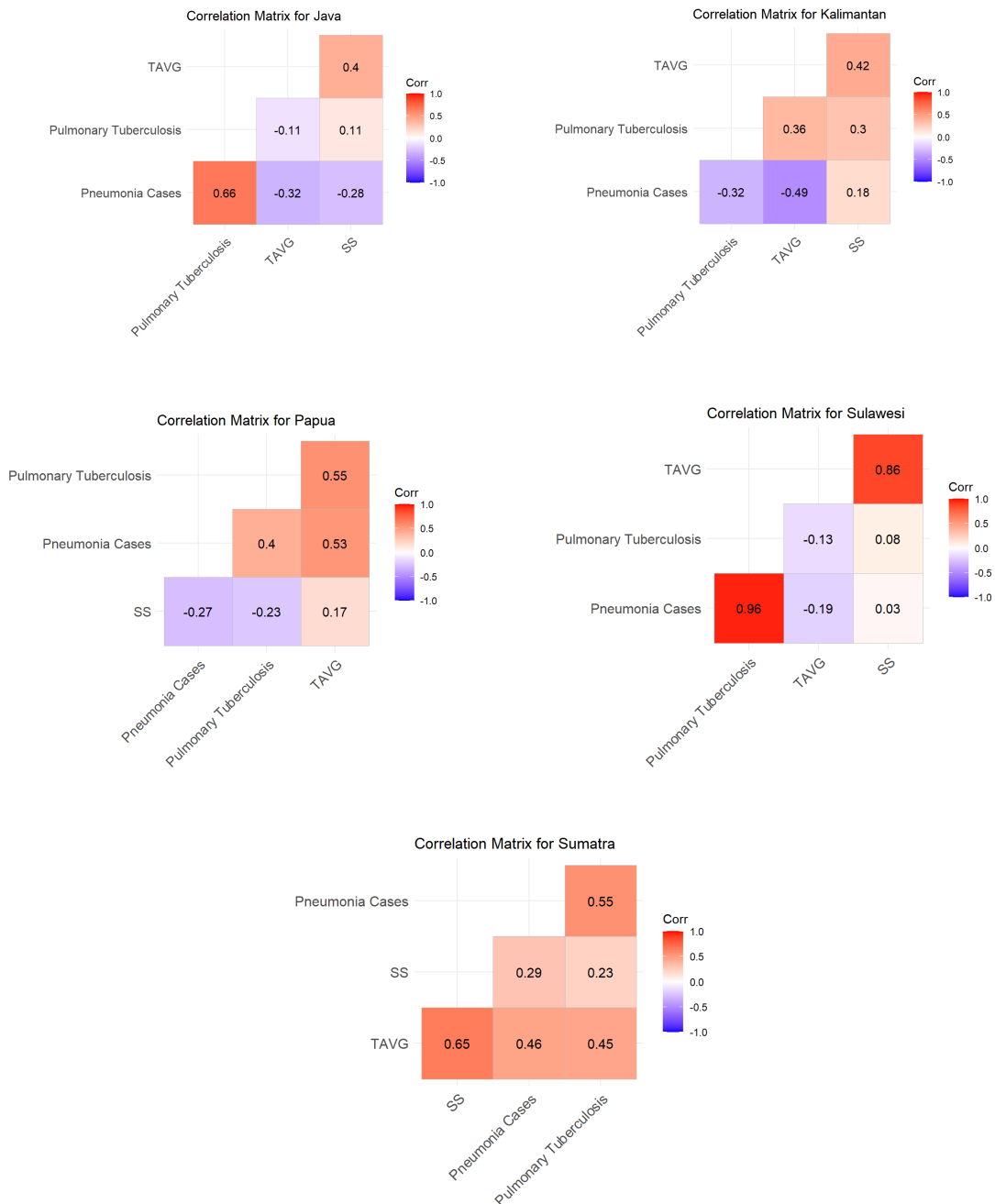


#1. Island-Specific Correlation Matrices



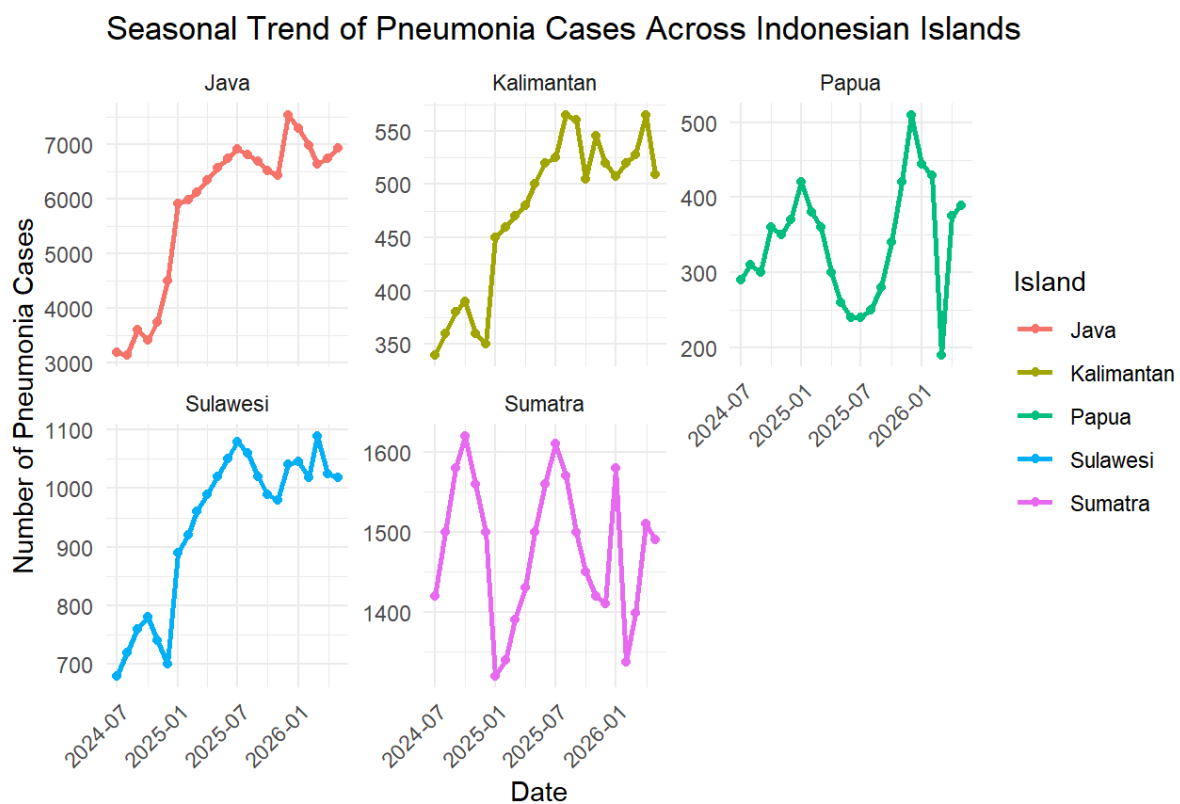
The key findings from the Island-Specific Correlation Matrices focus on how Air Quality and climate metrics interact with Respiratory Diseases (Pneumonia and TB) for each island:

- **Variable Relationships:** The matrices visualize the interaction between health metrics (Pneumonia_Cases, Pulmonary_Tuberculosis) and climate metrics (TAVG, RH_AVG, SS).
- **Correlation Patterns:**
 - **Positive (Warm/Red colors):** Indicates that as one variable increases, the other increases as well.

- **Negative (Cool/Blue colors):** Indicates that as one variable increases, the other decreases.
- **No Correlation (White/Faded colors):** Suggests the variables do not have a meaningful impact on each other.
- **Interpretation of Values:** The correlation strength is measured on a scale from -1.0 to 1.0. A high absolute value (e.g., 0.8) indicates a strong correlation, while values near 0 (e.g., -0.2) suggest a weak relationship.

The goal of this analysis is to provide a granular view of how climate factors correlate with disease burden independently for Java, Sumatra, Kalimantan, Sulawesi, and Papua.

#2. Spatio-Temporal Seasonal Trend Analysis



Based on the faceted time-series plot, breaking down the pneumonia data by island reveals that while some regions are driven by clean, predictable weather seasons, others are experiencing massive, macro-level surges.

1. The 2025 "Surge" Islands (Java, Sulawesi, Kalimantan)

Instead of a flat, repeating seasonal pattern, Java, Sulawesi, and Kalimantan all share a distinct macro-trend: a massive, non-seasonal step-up in case baselines that took place throughout 2025.

- Java: Cases practically rocketed from a baseline of ~3,500 in late 2024 to an all-time peak of over 7,500 cases by late 2025/early 2026.
- Sulawesi & Kalimantan: Both islands show nearly identical trajectories. They hit a minor trough around December 2024/January 2025, followed by a steep, year-long climb that peaked in mid-2025 and spiked again in early 2026.
- Insight: The sheer scale of this growth suggests an underlying factor beyond normal weather cycles—such as an epidemic wave, increased testing, or changes in regional healthcare reporting in 2025.

2. Papua's Textbook Monsoon Seasonality

Papua stands out as the cleanest example of classic meteorological seasonality. It follows a highly predictable, cyclical wave that aligns with Indonesia's wet and dry seasons:

- The Peaks: Clear, sharp peaks occur right around January 2025 (~420 cases) and January 2026 (~510 cases). This perfectly matches the height of the northwest monsoon (rainy season), when respiratory infections typically spike.
- The Troughs: Deep drops occur during the dry mid-year months, hitting a low around July 2025 (~240 cases) and an incredibly sharp plunge in early 2026 (~190 cases).

3. Sumatra's Counter-Cyclical Volatility

Sumatra plays by an entirely different set of rules compared to the other islands. It displays a highly volatile, oscillating pattern:

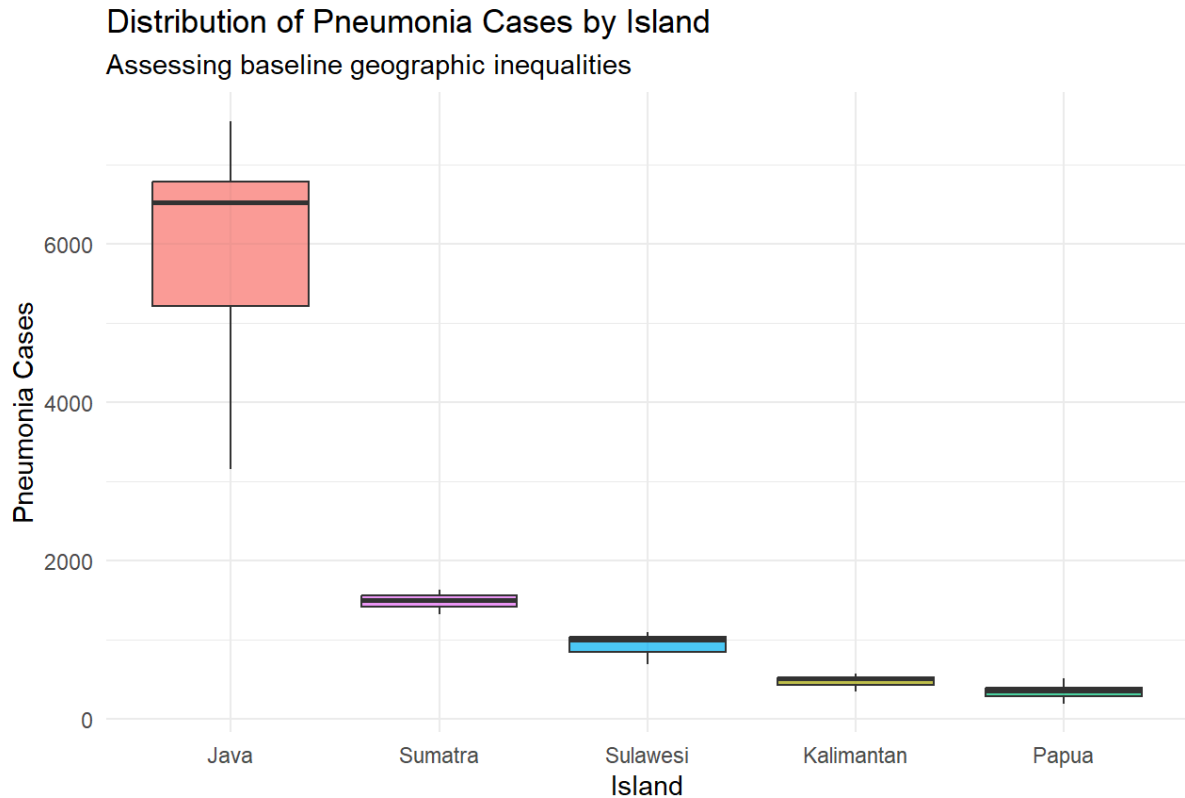
- An Inverse Pattern: While Papua peaks in January, Sumatra does the exact opposite. It experienced sharp, dramatic crashes in cases directly around January 2025 and January 2026.
- The Peaks: Sumatra's cases peak during the transition months (around October 2024 and mid-2025), bounded tightly between a narrow band of 1,350 to 1,620 cases.

4. Visual Validation of the ANOVA Scale Disparity

This seasonal chart perfectly explains *why* the previous ANOVA box plot had such massive variances:

- Java is operating on an entirely different axis (3,000 to 7,500 cases). Its massive vertical surge across 2025 is what caused its box plot to look so incredibly stretched out.
- By contrast, islands like Kalimantan (350–550 cases) and Papua (200–500 cases) look highly variable on their individual mini-charts, but their actual case volumes are a drop in the bucket compared to Java's overwhelming burden.

#3. Baseline Inequalities: Comparative Variance Analysis (ANOVA)



Based on the box plot provided, which lays the groundwork for an ANOVA (Analysis of Variance) test to assess geographic inequalities.

1. Extreme Geographic Disparity (Stark Visual Evidence for ANOVA)

There is a massive, visually obvious difference in the mean/median pneumonia cases across the islands.

- **Java is a massive outlier:** The entire box plot for Java sits entirely above the rest of the islands, with a median of roughly **6,500 cases**.
- **The Rest of the Islands:** Sumatra, Sulawesi, Kalimantan, and Papua all cluster much lower, with medians tracking under 2,000 cases.
- **ANOVA Expectation:** Because the between-group variance (the gaps between the islands) is so enormous compared to the within-group variance of the smaller islands, a standard ANOVA test would yield a **highly statistically significant p-value ($p < 0.05$)**, confirming that geographic location heavily influences case numbers.

2. Clear Hierarchical Step-Down Trend

The data shows a strict, descending burden of pneumonia cases from west to east/less populated islands:

- **Java:** ~6,500 cases (Highest)

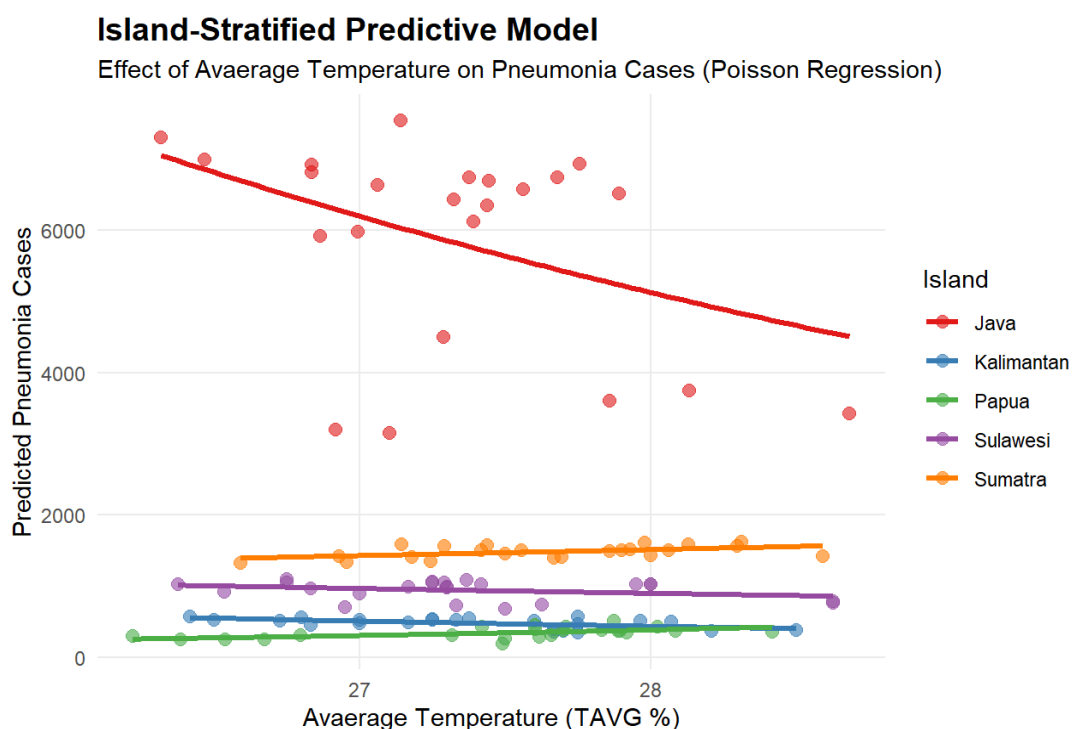
- **Sumatra:** ~1,500 cases
- **Sulawesi:** ~1,000 cases
- **Kalimantan:** ~500 cases
- **Papua:** ~300 cases (Lowest)

3. Violation of ANOVA Assumptions (A Critical Statistical Insight)

If you are running a standard One-Way ANOVA on this data, this chart flags a major technical issue: **Heteroscedasticity** (Unequal Variances).

- **Java's spread** (from roughly 3,200 to 7,500 cases) is incredibly wide, indicating high volatility or seasonality in case numbers.
- **The other four islands** have incredibly tight, narrow distributions (very low variance).
- **Statistical Fix:** Because standard ANOVA assumes equal variances across all groups, you should likely use a **Welch's ANOVA** (which doesn't assume equal variances) and pair it with a **Games-Howell post-hoc test** instead of a standard Tukey's HSD to find exactly which islands differ from one another.

#4.Island-Stratified Predictive Modeling



Based on the scatter plot and the island-stratified Poisson regression lines, this predictive model explores how average temperature (TAVG) influences pneumonia case counts.

1. Java's High Sensitivity to Temperature (Strong Negative Relationship)

Java is the only island where temperature acts as a dramatic predictor of pneumonia cases.

- The Trend: The red regression line shows a steep negative slope. As the average temperature increases from roughly 26°C to 28.5°C, the predicted pneumonia cases drop sharply from over 7,000 down to around 4,500.
- Implication: Lower temperatures in Java are strongly associated with higher predicted pneumonia burdens. This aligns with the rainy/monsoon season trends where cooler, wetter months typically see spikes in respiratory illness.

2. Temperature "Inelasticity" Across the Remaining Islands

For the other four islands, temperature fluctuations have an incredibly weak or negligible impact on predicting pneumonia cases.

- Flat Lines: Sumatra (orange) shows a very slight positive slope, while Sulawesi (purple), Kalimantan (blue), and Papua (green) feature almost entirely flat regression lines.
- Implication: Within this specific temperature bracket (roughly 26°C to 29°C), changing temperatures do not meaningfully drive or predict case variations in these regions. Other localized factors (like humidity, population density, or healthcare access) likely carry far more weight in a predictive model for these islands.

3. Dominance of Regional Intercepts over Slopes

The model confirms that where a patient is matters drastically more than the weather at any given time.

- The vertical separation between the regression lines (the "intercepts" in a Poisson model) perfectly mirrors the baseline geographic inequality seen in your ANOVA analysis.
- Even at Java's warmest, lowest-predicted point (~4,500 cases), its pneumonia burden remains miles above Sumatra's peak (~1,500 cases).

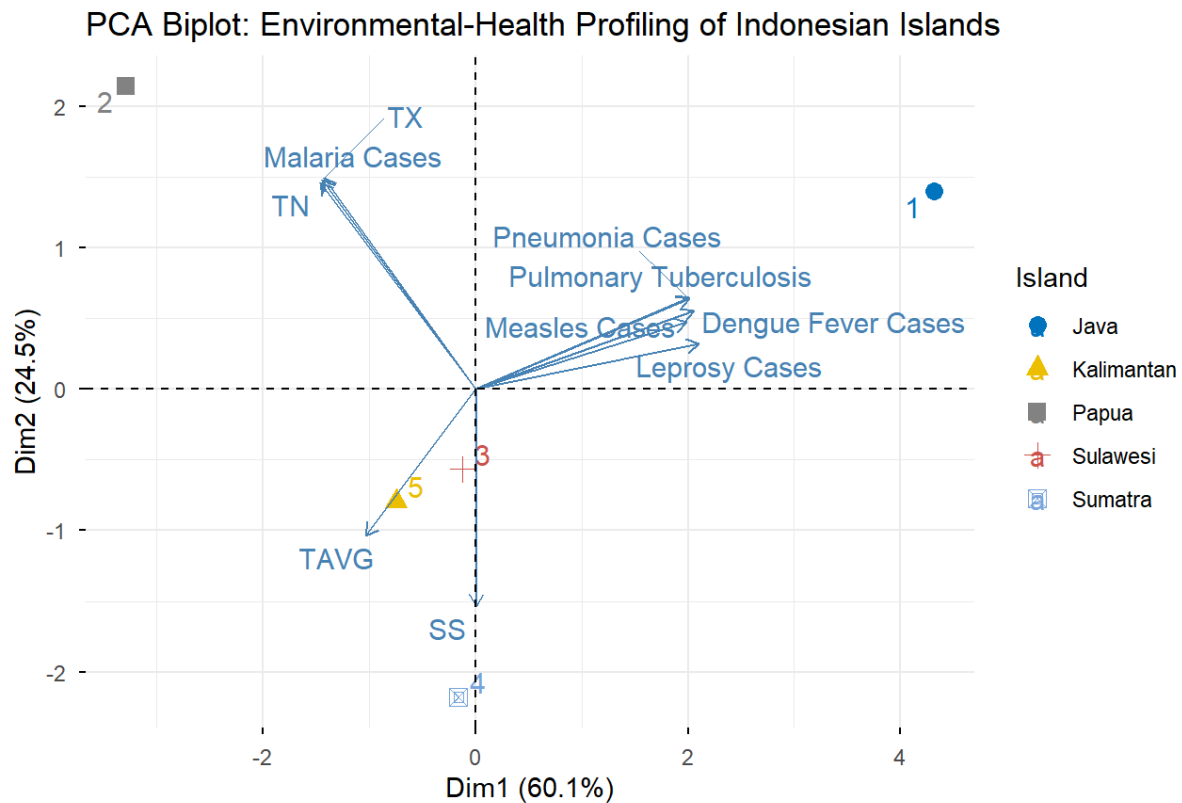
4. High Data Dispersion in the Java Model

Looking closely at the distribution of the individual data points:

- The data points for Sumatra, Sulawesi, Kalimantan, and Papua hug their respective regression lines very tightly.
- Java's data points, however, are highly scattered. There is a cluster of points remaining high (>6,000 cases) across all temperatures, alongside a few sudden drops below 4,000 cases.

- Statistical Insight: This high variance suggests that while temperature is a statistically significant predictor for Java, a simple single-variable Poisson model leaves a lot of unexplained noise. To improve predictive accuracy for Java, the model would benefit from adding interaction terms or additional features (like a time variable to account for that massive 2025 macro-surge).

#5. PCA Clustering of Island Environmental-Health Profiles



This biplot neatly categorizes the islands into three distinct health-environmental ecosystems:

1. Java: High-density, multi-disease hotspot (Respiratory/Infectious/Dengue).
2. Papua: Severe Malaria hotspot driven by temperature extremes (TX/TN).
3. Sumatra & Kalimantan: Climate-dominated zones defined by steady sunshine and average temperatures rather than epidemic-scale case counts.